

Homework Exercise Sheet

Ex. 10 Solution

Label (S) : only one correct answer; (M) : more than one correct answer

1. (S) Compute the first three iterations of the Newton method for $F(x) = x^3 + 2x^2 - 3x + 1$. Utilize the initial guess $x^{(0)} = -10$. Choose the correct value of $x^{(3)}$. Round intermediate computations to a single decimal point.

a) -3.9 b) 5.1 c) -4.2 d) 1.1

Solution:

a). Use the formula $x^{(k+1)} = x^{(k)} - F(x^{(k)})/F'(x^{(k)})$. $x^{(1)} = -7$, $x^{(2)} = -5.1$, $x^{(3)} = -3.9$.

2. (S) Compute the first two iterations of the *secant method* for $F(x) = x^3 + 3x^2 - 2x + 1$. Utilize the initial guesses $x^{(0)} = -10$, $x^{(1)} = -5$. Choose the correct value of $x^{(3)}$. Round intermediate computations to a single decimal point.

a) 3.0 b) -4.0 c) 4.1 d) -3.1

Solution:

b). Use the formula $x^{(k+1)} = x^{(k)} - (x^{(k)} - x^{(k-1)})F(x^{(k)})/(F(x^{(k)}) - F(x^{(k-1)}))$. $x^{(2)} = -4.7$, $x^{(3)} = -4.0$.

3. (S) Compute one step of the Newton method for the system $F(x, y) = 0$, where $F(x, y) = [xy^2 - 2, x^2 + y^2 - 3]^T$. Utilize the initial guess $x^{(0)} = y^{(0)} = 2$. Choose the correct values of $(x^{(1)}, y^{(1)})$.

a) (2, 0.75) b) (1, 0.75) c) (2, 1.75) d) (1, 1.75)

Solution:

d). Use the formula $z^{(k+1)} = z^{(k)} - DF(z^{(k)})^{-1}F(z^{(k)})$, where $z = [x, y]^T$ and DF is the Jacobian of F . Introducing $f_1(x, y) = xy^2 - 2$ and $f_2(x, y) = x^2 + y^2 - 3$, $DF(x, y) = \begin{bmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{bmatrix} = \begin{bmatrix} y^2 & 2xy \\ 2x & 2y \end{bmatrix}$. $DF(x, y)^{-1} = \frac{1}{2y^3 - 4x^2y} \begin{bmatrix} 2y & -2xy \\ -2x & y^2 \end{bmatrix}$.

4. (S) Select the correct formula for the Jacobian of $F : \mathbb{R}^{n,n} \rightarrow \mathbb{R}$, $F(\mathbf{X}) := \text{tr}(\text{inv}(\mathbf{X}))$ where $\text{tr} \mathbf{A} := \sum_{i=1}^n (\mathbf{A})_{ii}$ and $\text{inv} \mathbf{A} := \mathbf{A}^{-1}$.

- a) $DF(\mathbf{X}) = \{\mathbf{H} \in \mathbb{R}^{n,n} \mapsto -\mathbf{X}^{-1}\mathbf{H}\mathbf{X}^{-1}\}$
- b) $DF(\mathbf{X}) = \{\mathbf{H} \in \mathbb{R}^{n,n} \mapsto -\mathbf{X}^{-\top}\mathbf{H}\mathbf{X}^{-1}\}$
- c) $DF(\mathbf{X}) = \{\mathbf{H} \in \mathbb{R}^{n,n} \mapsto -\mathbf{X}^{-1}\mathbf{H}\mathbf{X}^{-\top}\}$
- d) $DF(\mathbf{X}) = \{\mathbf{H} \in \mathbb{R}^{n,n} \mapsto \text{tr}(-\mathbf{X}^{-1}\mathbf{H}\mathbf{X}^{-1})\}$
- e) $DF(\mathbf{X}) = \{\mathbf{H} \in \mathbb{R}^{n,n} \mapsto \text{tr}(-\mathbf{X}^{-\top}\mathbf{H}\mathbf{X}^{-1})\}$
- f) $DF(\mathbf{X}) = \{\mathbf{H} \in \mathbb{R}^{n,n} \mapsto \text{tr}(-\mathbf{X}^{-1}\mathbf{H}\mathbf{X}^{-\top})\}$

Remark: By $\mathbf{A}^{-\top}$ we mean $(\mathbf{A}^{-1})^{\top}$.

Solution:

d). F is the composition of $\text{tr} : \mathbb{R}^{n,n} \rightarrow \mathbb{R}$ and $\text{inv} : \mathbb{R}^{n,n} \rightarrow \mathbb{R}^{n,n}$. First, we figure out that the Jacobian of each mapping is $D\text{tr} = \{\mathbf{H} \mapsto \text{tr}\mathbf{H}\}$ and $D\text{inv}(X) = \{\mathbf{H} \mapsto -\mathbf{X}^{-1}\mathbf{H}\mathbf{X}^{-1}\}$ (see the derivation in Example 8.5.1.35). Then, by the chain rule $(D(F \circ G)(\mathbf{X}) = DF(G(\mathbf{X})) \circ DG(\mathbf{X}))$, it yields that $DF(\mathbf{X})$ is the linear mapping $\{\mathbf{H} \in \mathbb{R}^{n,n} \mapsto \text{tr}(-\mathbf{X}^{-1}\mathbf{H}\mathbf{X}^{-1})\}$.

5. **(M)** Given a nonlinear equation $F(\mathbf{X}) = \mathbf{0}$ where $F : \mathbb{R}^{n,n} \rightarrow \mathbb{R}^n$ is sufficiently smooth. Which of the following statements regarding Newton's method is **wrong**?
- a) Newton's method leads to the fixed-point iteration $\mathbf{X}^{(k+1)} = \mathbf{X} + DF(\mathbf{X}^{(k)})^{-1}F(\mathbf{X}^{(k)})$.
 - b) $DF(\mathbf{X})$ can be represented by an $(n \times n)$ -matrix.
 - c) In general, Newton's method is locally (at least) quadratically convergent.
 - d) If $DF(\mathbf{X}^*)$ and $\text{Hess}F(\mathbf{X}^*)$ are vanishing at the solution $\mathbf{X}^*$, Newton's method enjoys local cubic convergence.
 - e) The computational effort of each step of Newton's iteration is in general $\mathcal{O}(n^2)$.

Solution:

a), b), d), e). The right formula for Newton's iteration is $\mathbf{X}^{(k+1)} = \mathbf{X} - DF(\mathbf{X}^{(k)})^{-1}F(\mathbf{X}^{(k)})$. $DF(\mathbf{X})$ is supposed to be a linear mapping from $\mathbb{R}^{n,n}$ to $\mathbb{R}^n$ and thus is represented by an $(n \times n \times n)$ -matrix (or usually called "tensor"). The local quadratic convergence in 1D is proved in § 8.4.2.5 and it also applies to multiple dimensions. Statement d) should not be confused with Lemma 8.3.2.15 which states higher-order local convergence provided that the first several orders of the derivatives of the iteration function Φ are vanishing. In other words, Statement c) is true if F is replaced by Φ . Statement e) is wrong since the matrix inversion costs the computational effort of $\mathcal{O}(n^3)$ in general.

6. **(S)** Consider solving the equation $F(x) = 0, x \in \mathbb{R}$ with a multi-point method which approximate the function $F(x)$ with **piece-wise linear functions** defined by $(x^{(k)}, F(x^{(k)})), (x^{(k-1)}, F(x^{(k-1)})), (x^{(k-2)}, F(x^{(k-2)}))$. What is the resulting iteration formula using this method? Denote x_1, x_2, x_3 to be a permutation of $x^{(k)}, x^{(k-1)}, x^{(k-2)}$ such that $x_1 \leq x_2 \leq x_3$. Assume the piece-wise linear function always has a root.

$$\begin{aligned}
\text{a) } x^{(k+1)} &= \frac{F(x_1)x_2 - F(x_2)x_1}{F(x_1) - F(x_2)} \\
\text{b) } x^{(k+1)} &= \frac{F(x_2)x_3 - F(x_3)x_2}{F(x_2) - F(x_3)} \\
\text{c) } x^{(k+1)} &= \frac{F^2(x_1)(F(x_2)x_3 - F(x_3)x_2) + F^2(x_2)(F(x_3)x_1 - F(x_1)x_3) + F^2(x_3)(F(x_1)x_2 - F(x_2)x_1)}{F^2(x_1)(F(x_2) - F(x_3)) + F^2(x_2)(F(x_3) - F(x_1)) + F^2(x_3)(F(x_1) - F(x_2))} \\
\text{d) } x^{(k+1)} &= \begin{cases} \frac{F(x_1)x_2 - F(x_2)x_1}{F(x_1) - F(x_2)}, & \text{if } \frac{F(x_1)x_2 - F(x_2)x_1}{F(x_1) - F(x_2)} \leq x_2 \\ \frac{F(x_2)x_3 - F(x_3)x_2}{F(x_2) - F(x_3)}, & \text{otherwise} \end{cases}
\end{aligned}$$

Solution:

d). Since the piece-wise linear approximation is a combination of linear functions, the resulting iteration formula must be a mixture of the scant formula defined by $(x_1, F(x_1))$, $(x_2, F(x_2))$ and $(x_2, F(x_2))$, $(x_3, F(x_3))$. Therefore, we are selecting between $x_L^{(k+1)} = \frac{F(x_1)x_2 - F(x_2)x_1}{F(x_1) - F(x_2)}$ and $x_R^{(k+1)} = \frac{F(x_2)x_3 - F(x_3)x_2}{F(x_2) - F(x_3)}$. Due to the nature of the piece-wise linear function, $x_L^{(k+1)}$ is only valid when $x_L^{(k+1)} \leq x_2$, likewise, $x_R^{(k+1)}$ is only valid when $x_L^{(k+1)} \geq x_2$. According to assumption, one of the two has to be valid if the piece-wise linear function has a root. When both are valid, we just arbitrarily select $x_L^{(k+1)}$. In the end, the overall iteration formula should be

$$x^{(k+1)} = \begin{cases} x_L^{(k+1)}, & \text{if } x_L^{(k+1)} \leq x_2 \\ x_R^{(k+1)}, & \text{otherwise} \end{cases}$$

Remark: Compare this method with the bisection method.

7. **(S)** Given a nonlinear equation $F(\mathbf{X}) = \mathbf{0}$, $F: \mathbb{R}^m \rightarrow \mathbb{R}^n$, $m > n$, select the Newton's iteration for solving this equation.

Hint: This kind of problems in general has infinite solutions since inputs have more dimensions than outputs ("number of equations is less than number of unknowns"). We aim to find just one solution.

- Newton's method is not applicable because $DF \in \mathbb{R}^{n \times m}$ is not invertible.
- $\mathbf{X}^{(k+1)} = \mathbf{X}^{(k)} - DF(\mathbf{X}^{(k)})^{-1}F(\mathbf{X}^{(k)})$
- $\mathbf{X}^{(k+1)} = \mathbf{X}^{(k)} - DF(\mathbf{X}^{(k)})^\dagger F(\mathbf{X}^{(k)})$ where $\mathbf{A}^\dagger$ denotes the Moore-Penrose pseudoinverse of matrix $\mathbf{A} \in \mathbb{R}^{n \times m}$.
- $\mathbf{X}^{(k+1)} = \mathbf{X}^{(k)} - (DF(\mathbf{X}^{(k)})^\top DF(\mathbf{X}^{(k)}))^{-1}F(\mathbf{X}^{(k)})$

Solution:

c). We approximate $F(\mathbf{X})$ at $\mathbf{X}^{(k)}$ using formula 8.5.1.4:

$$\tilde{F}(\mathbf{X}) = F(\mathbf{X}^{(k)}) + DF(\mathbf{X}^{(k)})(\mathbf{X} - \mathbf{X}^{(k)}), \mathbf{X}^{(k)} \in \mathbb{R}^m.$$

However it is not possible to directly solve for $\tilde{F}(\mathbf{X}^{(k+1)}) = 0$ because $DF \in \mathbb{R}^{n \times m}$ is not invertible. However, since $\tilde{F}(\mathbf{X}^{(k+1)}) = 0$ is a linear equation, we can solve the **generalized least squares solution** (see Definition 3.1.3.1): $\mathbf{X}^{(k+1)} - \mathbf{X}^{(k)} = \text{lsq}(DF(\mathbf{X}^{(k)}), -F(\mathbf{X}^{(k)}))$

which leads to c).